Math 142, F19

Quiz 6

Name: Answer

Instructions: **You must show all of your work to receive credits for correct answers.**

Problem 1. (5 points) Use the trapezoidal rule and Simpson's rule respectively with $n = 4$ to eatimate the integral $\int_0^2 x^2 + 1\,dx$.

$a=0,\ b=2,\ n=4 \Rightarrow \Delta x = \frac{b-a}{n} = \frac{2-0}{4} = \frac{1}{2}$

i	x_i	$y_i = f(x_i)$
0	0	1
1	$\frac{1}{2}$	$\frac{5}{4}$
2	1	2
3	$\frac{3}{2}$	$\frac{13}{4}$
4	2	5

Trapezoidal Rule:

$$\int_0^2 x^2+1\,dx \approx T = \frac{\Delta x}{2}(y_0+2y_1+2y_2+2y_3+y_4)$$
$$= \frac{1}{2}\times\frac{1}{2}\left[1+2\left(\tfrac{5}{4}\right)+2(2)+2\left(\tfrac{13}{4}\right)+5\right]$$
$$= \frac{1}{4}\times 19 = \frac{19}{4}$$

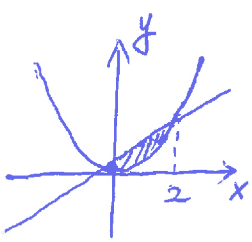
Simpson's Rule

$$\int_0^2 x^2+1\,dx \approx S = \frac{\Delta x}{3}(y_0+4y_1+2y_2+4y_3+y_4)$$
$$= \frac{1}{3}\times\frac{1}{2}\left[1+4\left(\tfrac{5}{4}\right)+2(2)+4\left(\tfrac{13}{4}\right)+5\right]$$
$$= \frac{1}{6}\times 28 = \frac{14}{3}$$

Exact Integration:

$$\int_0^2 x^2+1\,dx = \left[\frac{x^3}{3}+x\right]_0^2$$
$$= \left(\frac{8}{3}+2\right)-0$$
$$= \boxed{\frac{14}{3}}$$

Problem 2. (5 points) Find the area of the region bounded by the curves $y = 2x$ and $y = x^2$.



$$\begin{cases} y=2x \\ y=x^2 \end{cases} \Rightarrow x^2=2x \Rightarrow x^2-2x=0 \Rightarrow x(x-2)=0$$

$x=0$ or $x=2$

$\Downarrow \qquad\qquad \Downarrow$

$y=0 \qquad y=2$

$$\text{Area} = \int_0^2 (2x-x^2)\,dx$$
$$= \left[x^2-\frac{x^3}{3}\right]_0^2$$
$$= \left(4-\frac{8}{3}\right)-0 = \frac{4}{3}$$